

Desktop-to-Jetson Internet Forwarding and SICK LiDAR Network Recovery Guide

Persistent setup for two desktop Wi-Fi adapters, Jetson ROS networking, Docker host networking, and a dedicated SICK LiDAR subnet.

Device	Interface / Profile	Address / Role
Desktop	wlp70s0 / eduroam	Internet uplink (WAN)
Desktop	wlx289401d4e6cb / Robot_Network_5G	192.168.0.20/24 - robot network
Jetson	enx9c69d39444e8 / Wired connection 3	192.168.0.10/24 - gateway 192.168.0.20
Jetson	eno1 / Wired connection 1	192.168.131.1/24 - LiDAR only
SICK LiDAR	Ethernet	192.168.131.20:2112

Golden rule: eno1 is LiDAR-only and must never become the Jetson default route. The Jetson default route must point to the desktop at 192.168.0.20.

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1. Target Topology

```
Internet / eduroam
|
Desktop wlp70s0
|
Desktop forwarding + NAT
|
Desktop wlx289401d4e6cb = 192.168.0.20/24
|
Robot_Network_5G router / robot LAN
|
Jetson enx9c69d39444e8 = 192.168.0.10/24
|
Default gateway on Jetson = 192.168.0.20

Separate LiDAR path:
Jetson eno1 = 192.168.131.1/24
|
SICK LiDAR = 192.168.131.20:2112
```

The desktop routes only Jetson internet traffic from the robot network to eduroam. The LiDAR subnet remains completely separate.

2. Desktop Wi-Fi Profiles

Run this section on the desktop.

2.1 Bind eduroam to the internal Wi-Fi adapter

```
sudo nmcli connection modify "eduroam" \
connection.autoconnect yes \
connection.autoconnect-retries 0 \
connection.interface-name wlp70s0
```

2.2 Bind the robot network to the USB Wi-Fi adapter

```
sudo nmcli connection modify "Robot_Network_5G" \
connection.autoconnect yes \
connection.autoconnect-retries 0 \
connection.interface-name wlx289401d4e6cb \
ipv4.method manual \
ipv4.addresses 192.168.0.20/24 \
ipv4.gateway "" \
ipv4.dns "" \
ipv4.never-default yes \
ipv6.method disabled
```

2.3 Activate both Wi-Fi profiles

```
sudo nmcli connection up "eduroam" ifname wlp70s0
sudo nmcli connection up "Robot_Network_5G" ifname wlx289401d4e6cb
```

2.4 Verify the desktop

```
nmcli -f NAME,TYPE,DEVICE connection show --active
ip -4 -br addr
ip route get 8.8.8.8
```

Expected:

```
eduroam          -> wlp70s0
Robot_Network_5G -> wlx289401d4e6cb
wlx289401d4e6cb  -> 192.168.0.20/24
Internet route   -> wlp70s0
```

A profile shown as Robot_Network_5G 1 is only a duplicate NetworkManager profile name. The real SSID remains Robot_Network_5G. Keep one clean profile bound to the USB adapter.

3. Permanent Desktop Forwarding and NAT

Run this section on the desktop.

3.1 Enable IPv4 forwarding permanently

```
echo 'net.ipv4.ip_forward=1' | \
sudo tee /etc/sysctl.d/99-jetson-internet-forward.conf

sudo sysctl --system
sysctl net.ipv4.ip_forward
```

Expected:

```
net.ipv4.ip_forward = 1
```

3.2 Create the forwarding script

```
sudo tee /usr/local/sbin/jetson-internet-forward.sh >/dev/null <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

WAN_IF="wlp70s0"
ROBOT_IF="wlx289401d4e6cb"
JETSON_IP="192.168.0.10/32"

sysctl -w net.ipv4.ip_forward=1 >/dev/null

iptables -t nat -C POSTROUTING \
  -s "$JETSON_IP" \
  -o "$WAN_IF" \
  -j MASQUERADE \
  2>/dev/null ||
iptables -t nat -A POSTROUTING \
  -s "$JETSON_IP" \
  -o "$WAN_IF" \
  -j MASQUERADE

iptables -C FORWARD \
  -i "$ROBOT_IF" \
  -o "$WAN_IF" \
  -s "$JETSON_IP" \
  -m conntrack \
  --ctstate NEW,ESTABLISHED,RELATED \
  -j ACCEPT \
  2>/dev/null ||
iptables -I FORWARD 1 \
  -i "$ROBOT_IF" \
  -o "$WAN_IF" \
  -s "$JETSON_IP" \
  -m conntrack \
  --ctstate NEW,ESTABLISHED,RELATED \
  -j ACCEPT

iptables -C FORWARD \
  -i "$WAN_IF" \
  -o "$ROBOT_IF" \
  -d "$JETSON_IP" \
  -m conntrack \
  --ctstate ESTABLISHED,RELATED \
  -j ACCEPT \
  2>/dev/null ||
iptables -I FORWARD 1 \
```

```
-i "$WAN_IF" \  
-o "$ROBOT_IF" \  
-d "$JETSON_IP" \  
-m conntrack \  
--ctstate ESTABLISHED,RELATED \  
-j ACCEPT  
EOF  
  
sudo chmod +x /usr/local/sbin/jetson-internet-forward.sh
```

3.3 Create the boot service

```
sudo tee /etc/systemd/system/jetson-internet-forward.service >/dev/null <<'EOF'
[Unit]
Description=Forward desktop eduroam internet to Jetson
Wants=network-online.target
After=network-online.target NetworkManager-wait-online.service docker.service ufw.service

[Service]
Type=oneshot
ExecStart=/usr/local/sbin/jetson-internet-forward.sh
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl daemon-reload
sudo systemctl enable --now jetson-internet-forward.service
```

3.4 Verify the desktop service and firewall rules

```
systemctl status jetson-internet-forward.service --no-pager

sudo iptables -t nat -S POSTROUTING | grep 192.168.0.10
sudo iptables -S FORWARD | grep 192.168.0.10
```

The service must show active (exited). The iptables commands should display one MASQUERADE rule and two forwarding rules for 192.168.0.10.

4. Persistent Jetson Network Profiles

Run this section on the Jetson. Restarting Wired connection 3 can interrupt SSH.

4.1 Robot network and desktop gateway

```
sudo nmcli connection modify "Wired connection 3" \
connection.autoconnect yes \
connection.autoconnect-retries 0 \
connection.interface-name enx9c69d39444e8 \
ipv4.method manual \
ipv4.addresses 192.168.0.10/24 \
ipv4.gateway 192.168.0.20 \
ipv4.dns "1.1.1.1 8.8.8.8" \
ipv4.ignore-auto-dns yes \
ipv4.never-default no \
ipv6.method disabled
```

4.2 Dedicated LiDAR network

```
sudo nmcli connection modify "Wired connection 1" \
connection.autoconnect yes \
connection.autoconnect-retries 0 \
connection.interface-name eno1 \
ipv4.method manual \
ipv4.addresses 192.168.131.1/24 \
ipv4.gateway "" \
ipv4.dns "" \
ipv4.never-default yes \
ipv6.method disabled
```

4.3 Activate both Jetson profiles

```
sudo nmcli connection up "Wired connection 1"

sudo nmcli connection down "Wired connection 3"
```

```
sudo nmcli connection up "Wired connection 3"
```

Reconnect to the Jetson at **192.168.0.10** if the SSH session drops.

5. Final Verification

Run these commands on the Jetson.

5.1 Verify routes

```
ip route
```

The important routes must be:

```
default via 192.168.0.20 dev enx9c69d39444e8
192.168.0.0/24 dev enx9c69d39444e8 src 192.168.0.10
192.168.131.0/24 dev eno1 src 192.168.131.1
```

5.2 Test gateway, internet, DNS, and LiDAR

```
ping -c 3 192.168.0.20
ping -c 3 8.8.8.8
getent hosts google.com
ping -c 3 192.168.131.20
nc -vz 192.168.131.20 2112
```

5.3 Verify SICK Docker networking

```
docker inspect -f '{{.HostConfig.NetworkMode}}' main-ws
```

Expected:

```
host
```

The SICK container must use **--network host**. Port mapping is not required because the driver initiates an outbound TCP connection to the LiDAR.

5.4 Test LiDAR from inside Docker

```
docker exec -it main-ws bash

ip route get 192.168.131.20
ping -c 3 192.168.131.20

timeout 3 bash -c '</dev/tcp/192.168.131.20/2112' \
  && echo "LiDAR port 2112 is reachable" \
  || echo "LiDAR port 2112 is NOT reachable"
```

5.5 Launch the ROS 1 SICK driver

```
roslaunch sick_scan_xd sick_lms_1xx.launch \
  hostname:=192.168.131.20 \
  port:=2112
```

The driver log must say it is connecting to **192.168.131.20:2112**. If it still shows **192.168.0.1**, inspect the launch file for an overridden hostname.

6. Recovery Checklist

Symptom	Check	Expected / Fix
Jetson cannot ping 8.8.8.8	ip route	Default route must be via 192.168.0.20, not 192.168.0.1.
Jetson can ping 8.8.8.8 but DNS fails	getent hosts google.com	Confirm Wired connection 3 DNS is 1.1.1.1 and 8.8.8.8.
Desktop forwarding disappears after reboot	systemctl status jetson-internet-forward.service	Enable service and verify it is active (exited).
Desktop robot Wi-Fi profile will not connect	nmcli connection show	Remove duplicate Robot_Network_5G profiles and recreate one bound to wlx289401d4e6cb.
LiDAR ping fails	ip -4 -br addr; ip route	eno1 must be 192.168.131.1/24 with no gateway.
LiDAR works on host but not Docker	docker inspect ... NetworkMode	Container must use host networking. Recreate it if needed.
SICK driver uses 192.168.0.1	Driver startup log / launch file	Pass hostname:=192.168.131.20 or fix the launch-file default.

7. Quick Status Commands

Desktop

```
nmcli -f NAME,TYPE,DEVICE connection show --active
ip -4 -br addr
ip route get 8.8.8.8
sysctl net.ipv4.ip_forward
systemctl status jetson-internet-forward.service --no-pager
sudo iptables -t nat -S POSTROUTING | grep 192.168.0.10
sudo iptables -S FORWARD | grep 192.168.0.10
```

Jetson

```
nmcli -f NAME,TYPE,DEVICE connection show --active
ip -4 -br addr
ip route
ping -c 3 192.168.0.20
ping -c 3 8.8.8.8
ping -c 3 192.168.131.20
nc -vz 192.168.131.20 2112
```

Do not change the LiDAR IP unless the scanner itself has been reconfigured. The working scanner address for this setup is 192.168.131.20 on TCP port 2112.